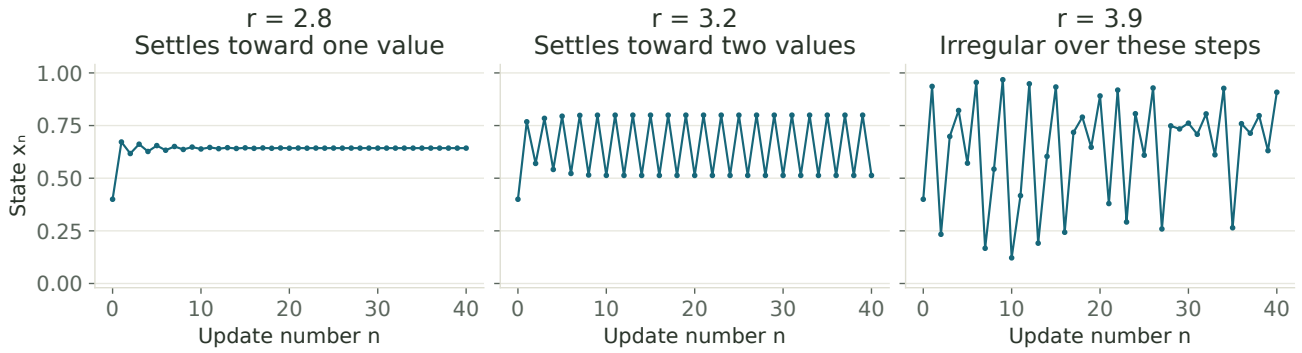
CHAPTER 10 / COMPLEXITY AND THE LIMITS OF PREDICTION

Recursion, fractals, and prediction

Follow a number through a maze, build a curve that keeps getting longer, and discover why an exact rule can still leave a forecast uncertain.

Repeated updates are already familiar: a balance earns interest; a controller acts on a temperature reading. This chapter builds on those examples from Weeks 1–9, using arithmetic, percentages, tables, and graph reading to investigate less familiar patterns.



Three patterns we will learn to read: settling toward one value, repeating between two values, and moving irregularly within fixed bounds. Each plot follows one value through repeated updates.

The question for the week

If we know the rule, do we know the future? Begin by defining the quantity, what one update means, the starting information, and the question we want the calculation to answer.

One rule can produce a long number trajectory, intricate geometry, or sensitivity to a starting measurement. We will distinguish calculation for chosen examples from the stronger conclusions that require proof.

Repeated updates show how a result develops. Calculating many particular cases helps us investigate a pattern, but it does not by itself establish a claim for every starting value.

Online reading · Week 10 explorer · Guided laboratory

10.1 Before trusting a forecast: state, rule, and question

Imagine a local conservation group deciding whether a fish-population forecast is reliable enough to guide its work next season. A rule takes an estimated current state and calculates a future state. Before acting, the group needs to know how the state was measured, which rule was chosen, and what the decision requires.

A *mathematical model* is a deliberately simplified description of a situation. We choose what to keep track of and write a rule relating the quantities. For example, a savings model might include interest and omit fees. A population model might include crowding and omit migration. A calculated answer describes what the model predicts under those assumptions.

Set up the mathematical world before calculating

1. *Choose the quantity.* A balance might be measured in dollars; a population might be measured in fish.
2. *Choose what one update means.* For a balance it could be one month. For a population it could be one breeding season. For a drawing, an update means one construction stage.
3. *Choose the starting information and the rule.* These tell us where to begin and how to obtain the next value.
4. *Choose the question.* Are we asking for the next value, a long-run pattern, the largest value, or the difference between two forecasts?

The word *state* means the current condition represented by the model: a balance, temperature, population, or other quantity. The *initial state* is its starting condition. *Dynamics* means how that state changes as we apply updates.

The population example is a simplified model of a real decision. Collatz and the fractal constructions are exact mathematical rules; they help isolate limits of computation and measurement without pretending that an integer sequence is literally a lake.

What do “complexity” and “chaos” mean here?

We will use *complexity* for complicated structures or behavior generated by comparatively simple rules. *Chaos* has a more specific mathematical meaning, involving deterministic dynamics and sensitive dependence on initial conditions. Long computations, unproved conjectures, and scale-dependent measurements pose different questions.

The calculations rely on arithmetic, percentages, tables, and graphs. The explorer provides longer trajectories that can be compared with calculations by hand.

10.2 How nearby starting estimates change

A *recursion* uses a current value to calculate the next value. A *trajectory* is the sequence of values it produces. The starting value has index 0; the result after one update has index 1.

Read the formula as an instruction

In $a_{n+1} = a_n + 5$, the letter a names the sequence. The small subscript tells us *which step* we are reading:

Symbol	Read it as	Meaning
n	update number	0 at the start; then 1, 2, 3, ...
a_n	"a at step n"	the current value
a_{n+1}	"a at the next step"	the new value after one more update
$a_{n+1} = a_n + 5$	"next value equals current value plus 5"	an instruction we can repeat

The subscript in a_0 identifies the starting value. Similarly, the $n + 1$ in the subscript labels the next step. The amount added to the current value is the 5 on the right side.

starting value $a_0 \longrightarrow a_1 \longrightarrow a_2 \longrightarrow \cdots$

Example · Read the subscript as a step counter

Imagine a community project with \$10,000 and an expected additional \$5,000 in donations each month. In thousands of dollars, start with $a_0 = 10$ and apply $a_{n+1} = a_n + 5$. The values are $10, 15, 20, 25, 30, 35$. The value after five updates is $a_5 = 35$. There are *six recorded values*, including the start.

Compare two nearby starts

Suppose two reports give the project's current balance as \$10,000 or \$10,100. Under the same \$5,000 monthly donation assumption, the starting numbers are 10 and 10.1 in thousands of dollars. Compare them after the same number of months. This simple rule is useful for introducing the difference between starting estimates; a real project would also have uncertain donations and expenses.

To describe this behavior, specify which difference is growing and over how many updates. We need to say which difference we mean.

A difference in units and a difference in percent

Suppose two model balances are \$50 and \$52. They are \$2 apart. Relative to the first balance, that gap is $2/50 = 0.04 = 4\%$.
Absolute difference asks "how many units apart?" *Relative difference* asks "how large is that gap compared with the reference value?" The bars in $|b_n - a_n|$ mean take the nonnegative size of the difference. For example, both $|52 - 50|$ and $|50 - 52|$ equal 2.

absolute difference $= |b_n - a_n|$ relative difference $= \dfrac{|b_n - a_n|}{|a_n|}$

The relative difference above uses the first trajectory as its reference and requires $a_n \neq 0$. Multiply by 100 to express it as a percentage. The condition on the reference says it cannot be zero, because division by zero is undefined.

Rule and starts	After the updates	What happened?
Add 5; start at 10 and 10.1	After 5 updates: 35 and 35.1	Absolute gap stays 0.1.
Multiply by 3; start at 1 and 1.01	After 6 updates: 729 and 736.29	Gap is 7.29, but relative gap stays 1%.
Multiply by 0.5; start at 10 and 10.1	After 5 updates: 0.3125 and 0.315625	Absolute gap shrinks to 0.003125.

A growing gap can still have a simple explanation

For an idealized message-sharing process, we might suppose that views triple each round before contacts begin to overlap. Under multiplication by 3, the gap also multiplies by 3 at each update. We can explain its growth exactly: $0.01 \cdot 3^n$. This is a useful comparison for later. It shows why "the forecasts are farther apart" needs an explanation of *how* they separate.

Pause and try 1

Start at 4 and 4.1. Double both values three times. What are the absolute and relative differences?

10.3 An exact rule can leave an open question

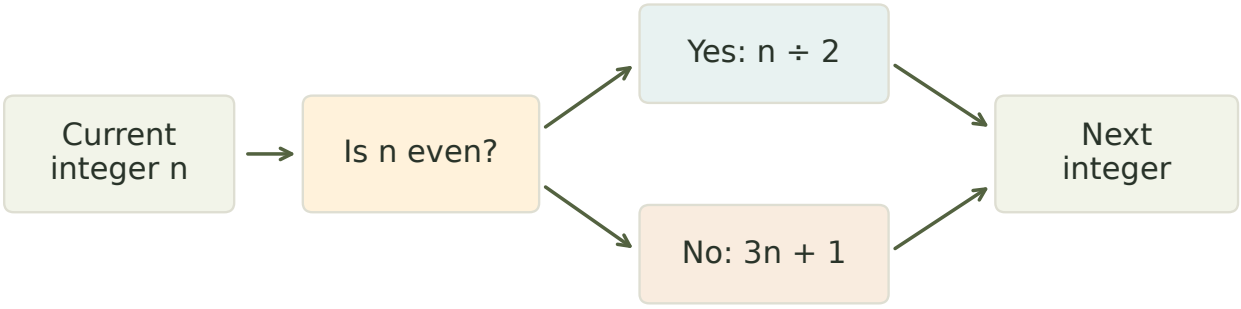
Our lake model involved an uncertain initial measurement. Now take an *exact* positive integer and follow a completely specified next-step rule. Even with perfect arithmetic, calculating examples does not necessarily answer a question about every possible starting integer.

Here the mathematical world is the positive integers: 1, 2, 3, and so on. We are not modeling fish or money yet. We are exploring what a particular arithmetic instruction does to a whole number. One update means applying the instruction once.

Choose a positive whole number. If it is even, divide it by 2. If it is odd, multiply it by 3 and add 1. Use the result as the next input.

An *even* integer divides by 2 without a remainder, such as 6 or 10. An *odd* integer does not, such as 3 or 5. The displayed formula below simply puts those two verbal instructions into one statement. The brace says "choose the appropriate row."

$$a_{n+1} = \begin{cases} a_n/2, & a_n \text{ is even,} \\ 3a_n + 1, & a_n \text{ is odd.} \end{cases}$$



Repeat with the new integer. Stop the recorded trajectory at its first 1.

The even/odd test chooses the branch. No coin is tossed; each next value is determined by the current integer.

Example · Trace the route from 6

First, 6 is even, so $6/2 = 3$. Then 3 is odd, so $3(3) + 1 = 10$. Continue by checking the *new* value each time.

$$6 \to 3 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1.$$

There are 8 arrows, so the trajectory first reaches 1 after *8 updates*. Its largest value, or *peak*, is 16. If we continue after 1, we get the repeating cycle $1 \to 4 \to 2 \to 1$.

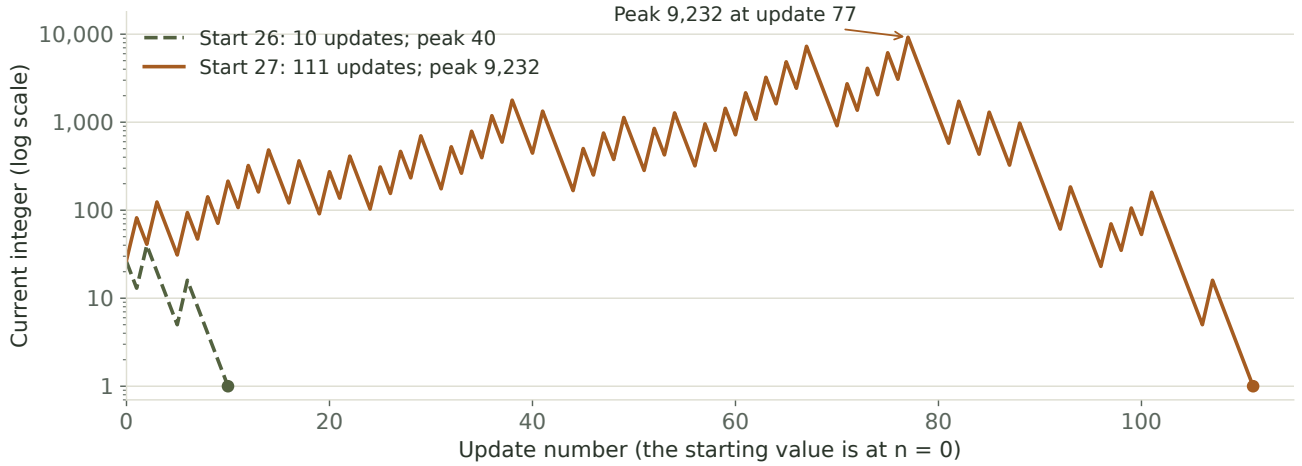
Nearby starts can take very different routes

Read this graph before comparing the routes

The horizontal axis counts updates: a point above 20 describes the value after 20 instructions. The vertical axis gives the current integer.

That vertical axis uses a *logarithmic scale*. Read its labels: 1, 10, 100, 1,000, 10,000. Moving up one equal-sized interval multiplies the value by 10. This lets a peak of 40 and a peak of 9,232 appear on the same graph. The labels, rather than the visual height alone, tell us the values.

First compare the horizontal endpoints to see which route lasts longer. Then use the vertical labels and the table to compare peaks.



Read the horizontal axis as an update count. On the logarithmic vertical axis, equal spacing represents multiplication by 10, so both the short and the tall trajectory remain visible.

Start	Updates to first 1	Peak
6	8	16
7	16	52
26	10	40
27	111	9,232

"Larger start" does not reliably mean "longer journey." For example, 32 reaches 1 in just five updates because $32 \to 16 \to 8 \to 4 \to 2 \to 1$. The explorer lets you trace an entire route and compare neighboring starts.

A connection to networks

Imagine each positive integer as a node, with an arrow to its next Collatz value. If two trajectories meet at the same number, their remaining routes agree exactly. Starting at 6 and 7, both routes eventually meet at 10. This is a directed network generated by a rule.

Pause and try 2

Starting at 5, record the trajectory to its first 1. How many updates are there? What is the peak?

10.4 How much do tested cases establish?

A *conjecture* is a proposed mathematical statement that has not yet been proved. A *proof* is an argument that covers every case described by the statement. The words "every positive integer" ask for much more than a large collection of successful examples.

The *Collatz conjecture* says that every positive integer eventually reaches 1. Extensive computer searches support the conjecture, but a proof for every positive integer is still unknown. The exact rule is easy to state; the universal claim is difficult to establish.

Example · A family we really can prove

Recall that a power counts repeated multiplications: $2^3 = 2 \cdot 2 \cdot 2 = 8$, and $2^4 = 16$. Starting from 8 gives $8 \to 4 \to 2 \to 1$; starting from 16 gives one extra halving first. The same reasoning works however many factors of 2 we start with.
If the starting number is a power of 2, say 2^k, each even step halves it:

$$2^k \to 2^{k-1} \to \cdots \to 2 \to 1.$$

It reaches 1 after exactly k updates. This argument works for *every* nonnegative integer k, so it covers infinitely many starts. It does not cover all positive integers: 7 and 27, for example, are not powers of 2.

A finite computation can settle a stated finite range. A universal proof must explain why no positive integer outside that range can escape. Conversely, a single positive integer proved never to reach 1 would refute the conjecture.

Two different questions about prediction

For any chosen Collatz integer, the next arithmetic step is straightforward. The unresolved question concerns every possible starting integer. In the population example earlier in this chapter, the rule was fixed while the initial count was uncertain. Here the starting integer and rule are exact, but a finite test does not settle a statement about every integer. These are different limits.

Pause and try 3

A program tests every starting integer from 1 through 10,000, and they all reach 1. Which statement is justified?

10.5 An intricate boundary can follow a simple rule

Suppose a restoration team needs to estimate how much shoreline to inspect or protect. First we use exact geometric constructions to see how the number of pieces, their lengths, and their areas change under repeated refinement. Then we return to the real issue of measuring an irregular bank.

Now one update changes a drawing rather than a single population or balance. Think of a drawing program with a replacement instruction. Stage 0 is the original drawing. Stage 1 applies the instruction once to every original piece. Stage 2 applies it once to every piece produced at stage 1.

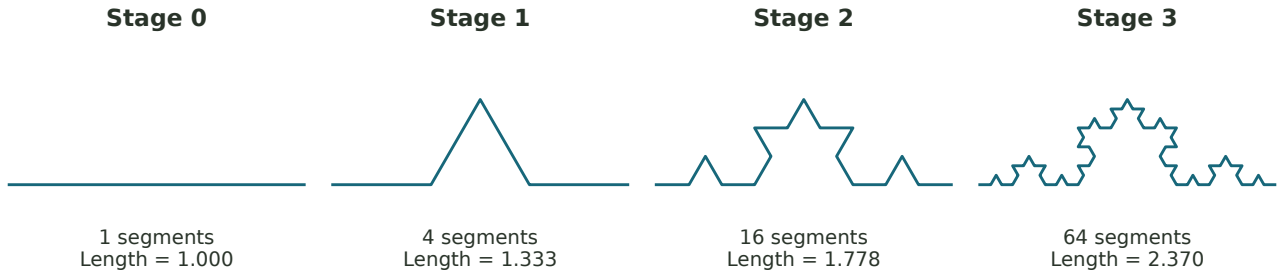
A self-similar object contains smaller copies of its overall pattern. We can build ideal fractals by repeating a geometric instruction at finer and finer scales. A displayed stage is a finite approximation; the ideal fractal is the limit of all stages.

For these examples, *self-similar* means a smaller part is an exact scaled copy of the whole limiting pattern. A *fractal* has structure that remains intricate as we inspect finer scales. We can draw only finitely many stages, but we can ask what the measurements do as the number of stages increases.

The Koch curve: more pieces, longer total

The distance straight across between two endpoints is different from the distance along a route between them. A winding trail can be much longer than the straight path with the same endpoints. The Koch construction makes that distinction visible and calculable.

1. Begin with a straight segment of length 1.
2. Divide it into thirds. Replace the middle third with the two sides of an equilateral triangular bump.
3. Apply that replacement to *every* segment of the new curve.



The endpoints stay the same distance apart. The route between them gets longer because each old segment becomes four segments, each one-third as long.

Two accounts to keep: how many, and how long?

After the first Koch replacement there are *four* pieces. Each piece is only *one-third* as long as the original. So four pieces do not make a total length of 4; their total is $4 \cdot (1/3) = 4/3$.
After the second replacement, each of those four pieces becomes four smaller pieces: $4 \cdot 4 = 16$ in all. Each new piece is one-third of one-third of the original: $(1/3)(1/3) = 1/9$. Total length is therefore $16 \cdot (1/9) = 16/9$.
The formulas below abbreviate these repeated multiplications. N_n counts segments, ℓ_n is the length of one segment, and L_n is the total length. The expression $(4/3)^n$ means multiply by $4/3$ once for

each stage.

$$N_n = 4^n, \qquad \ell_n = \left(\frac{1}{3}\right)^n, \qquad L_n = N_n \ell_n = \left(\frac{4}{3}\right)^n.$$

Stage n	Segment count N_n	Length per segment ℓ_n	Total length L_n
0	1	1	1
1	4	1/3	$4/3 \approx 1.3333$
2	16	1/9	$16/9 \approx 1.7778$
3	64	1/27	$64/27 \approx 2.3704$
4	256	1/81	$256/81 \approx 3.1605$

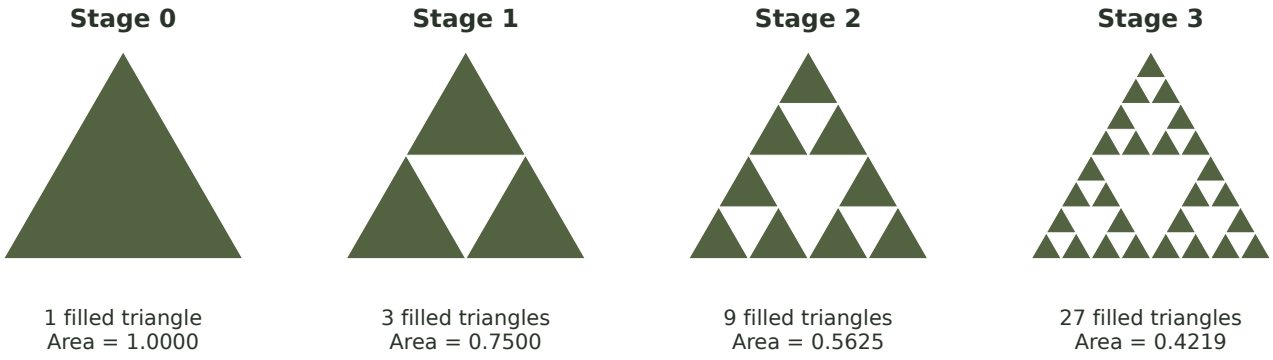
Example · A 9-meter starting segment

Imagine a nine-meter decorative border designed with a repeating zigzag. After one replacement there are four 3-meter segments, totaling 12 meters. After two replacements there are sixteen 1-meter segments, totaling 16 meters.
Using the formula gives the same result: $L_2 = 9(4/3)^2 = 16$ meters. The formula $(4/3)^n$ by itself assumes the original length is 1.

The Sierpinski triangle: more pieces, less area

Here we measure a filled region. Its *area* tells how much flat surface it covers, in square units. Keeping track of area differs from measuring the length of an edge.

Start with a filled triangle of area 1. Split it into four congruent triangles and remove the central one. Repeat on each remaining filled triangle. Each step keeps three quarters of the currently filled area.



The number of pieces multiplies by 3, but each piece has one-quarter the area of its parent. Count and area answer different questions.

Why does each small triangle have one-quarter the area?

Each small triangle has half the base and half the height of its parent. The area formula, $\frac{1}{2} \times$ base $\times$ height, is therefore multiplied by $(1/2)(1/2) = 1/4$. Keeping three such pieces keeps $3 \cdot (1/4) = 3/4$

of the parent's area.

Stage	Filled pieces	Area of each piece	Total remaining area
0	1	1	1
1	3	1/4	$3/4 = 0.75$
2	9	1/16	$9/16 = 0.5625$

For a concrete size, start with a triangle of area 16 square centimeters. After one stage, 12 square centimeters remain. After two, 9 remain. The *number* of pieces increases while the *total area* decreases.

$$T_n = 3^n, \qquad A_n = \left(\frac{3}{4}\right)^n.$$

At stage 4, there are $3^4 = 81$ filled triangles, yet only $(3/4)^4 = 81/256 \approx 0.3164$ of the original area remains. That is about 31.64%, not 0.3164%. In the ideal limit, the remaining area tends to zero while a rich self-similar set remains.

What does "in the limit" mean?

We never finish infinitely many drawing stages. Instead, we study the pattern in their measurements. Koch lengths are multiplied by a factor greater than 1, so they eventually exceed any chosen length. Sierpinski areas are multiplied by a positive factor less than 1, so they get as close to zero as we choose.

The ideal Sierpinski set can have infinitely many points and still cover zero square units. A straight line segment also contains infinitely many points while having zero area in the plane. Counting points, measuring length, and measuring area ask different questions.

Geometric complexity can be exactly predictable

The Koch curve acquires intricate detail, but its finite-stage counts and lengths have exact formulas. A fractal shape does not automatically imply an unpredictable time evolution.

Optional extension · What could a dimension between 1 and 2 mean?

A line segment can be made of two half-length copies of itself. A filled square can be made of four half-size copies: two across and two up. The number of copies needed at a smaller scale helps distinguish a line from a filled region.

A Koch curve has four copies at one-third scale. Four is more than the three copies needed to make a straight segment at that scale, but fewer than the nine third-size squares needed to cover a square. Its scaling falls between those familiar cases. The dimension numbers below describe this scaling relationship.

The logarithm formula records that relationship for readers who want it. You do not need to calculate logarithms for this week's laboratory or quiz. Here b is the reduction factor: $b = 3$ means each copy has one-third the original length.

For these exact self-similar examples, dimension compares the number of smaller copies with their scale. If there are N copies, each shrunk by a factor b, the similarity dimension satisfies $b^D = N$, so $D = \log N / \log b$.

Koch: $D = \log 4 / \log 3 \approx 1.262$. Sierpinski: $D = \log 3 / \log 2 \approx 1.585$. These dimensions describe

scaling; they are not percentages of the plane that the shape fills.

Pause and try 4

At Koch stage 3, does each segment have length 1/3 or 1/27? How does that produce the total length?

10.6 Predicting a population using a limited model

The conservation group wants to know when its survey and management plans should be revisited. The logistic map makes the update calculation clear, but it omits migration, habitat changes, age structure, and imperfect measurements. It is a mathematical demonstration, not a fitted forecast for any particular lake.

A small population model

Imagine monitoring a lake once per breeding season. Let 100,000 fish be a convenient reference count. A survey estimate of 20,000 fish becomes the fraction $20{,}000/100{,}000 = 0.2$, or 20% of that reference.

Writing a quantity on this common 0-to-1 scale is called *normalizing* it. In this model, $x = 0$ means zero fish, $x = 0.5$ means 50,000 fish, and $x = 1$ means 100,000 fish. A state of 0.2 represents 20,000 fish on this scale. The reference is a normalization choice, not a measured ecological carrying capacity.

To model a next season, we want growth to depend on the present population and to weaken when crowding is high. The logistic map is one simple rule built for that purpose. It is a mathematical example, with a chosen reference scale, rather than a complete ecological account of a pond.

The rule combines the current fraction with a growth setting and a limiting factor. In words: next fraction = growth setting × current fraction × remaining fraction.

We write the current fraction as x_n. The next season's fraction is x_{n+1}. The setting r is a *parameter*: a number we hold fixed during a run. Choosing a different r changes the model's response strength, much as changing the controller's strength changed the temperature sequence in Week 5.

In mathematics, a *map* is an input-output rule. The name *logistic map* refers to this particular population-style update. Repeating an update is also called *iterating* the rule.

$$x_{n+1} = r x_n (1 - x_n).$$

- x_n is the current state, normalized between 0 and 1.
- r is a fixed parameter controlling the update.
- $1 - x_n$ is a limiting factor: it gets smaller when x_n gets larger.

If $x_n = 0.2$, then $1 - x_n = 0.8$. If $x_n = 0.8$, then $1 - x_n = 0.2$. This complement is the limiting factor in the model: a larger current fraction leaves a smaller factor to multiply by.

Read r as a setting, not a percentage

In Week 6, multiplying a balance by 1.05 represented 5% growth each period. Here the whole multiplier is $r(1 - x_n)$, so it changes when x_n changes. For example, with $r = 2$:

Current fraction	Limiting factor	Whole multiplier	Next fraction
0.2	$1 - 0.2 = 0.8$	$2 \times 0.8 = 1.6$	$1.6 \times 0.2 = 0.32$
0.5	$1 - 0.5 = 0.5$	$2 \times 0.5 = 1$	$1 \times 0.5 = 0.5$
0.8	$1 - 0.8 = 0.2$	$2 \times 0.2 = 0.4$	$0.4 \times 0.8 = 0.32$

At a low population, the model's multiplier can exceed 1 and the next population grows. At a high population, the limiting factor can make the multiplier smaller than 1 and the population falls. The chosen rule can make a strong correction. This dependence on the current state makes the update *nonlinear*: its effects do not follow one fixed additive or multiplicative pattern.

For $0 \le r \le 4$, a starting state in $[0, 1]$ stays in that interval. We can use x_n as a simplified population index, but a real population requires more information than this one equation includes.

Example · Do not reuse the old input

Let $r = 2$ and $x_0 = 0.2$.

1. $x_1 = 2(0.2)(1 - 0.2) = 2(0.2)(0.8) = 0.32$.
2. $x_2 = 2(0.32)(1 - 0.32) = 2(0.32)(0.68) = 0.4352$.
3. $x_3 = 2(0.4352)(0.5648) = 0.49160192$.

Each new result replaces the previous input. The values approach 0.5.

In the 1,000-fish setting, those fractions represent approximately 200, 320, 435, and 492 fish. These are model estimates; we round fish counts for reporting while keeping the fractional calculation unrounded. A "generation" or "update" on the graph means one breeding season in this example.

A fixed point: an update that leaves the state alone

Try $r = 2$ and $x = 0.5$. The next value is $2(0.5)(1 - 0.5) = 0.5$. If the model reaches exactly 0.5, it keeps returning 0.5. This unchanged value is called a *fixed point*.

A sequence can also *approach* a fixed point: its later values get closer and closer to it. Our earlier run, $0.2, 0.32, 0.4352, 0.49160192, \ldots$, moves toward 0.5. A fixed point describes a lasting balance, whereas a first update describes what happens immediately from a particular start.

For the nonzero fixed point of this rule, we can use $x^* = 1 - 1/r$. The star is a label for the fixed-point value; it is not multiplication. To use the formula, divide 1 by the chosen r, then subtract that result from 1. For $r = 2.8$, this gives approximately 0.642857.

Why does the fixed-point formula work?

A fixed point has the same value before and after the update, so it satisfies $x = rx(1 - x)$. For $x \neq 0$, divide both sides by x, giving $1 = r(1 - x)$. Divide by r, then rearrange: $1/r = 1 - x$, so $x = 1 - 1/r$. Zero is also a fixed point: without any starting population, this model produces none.

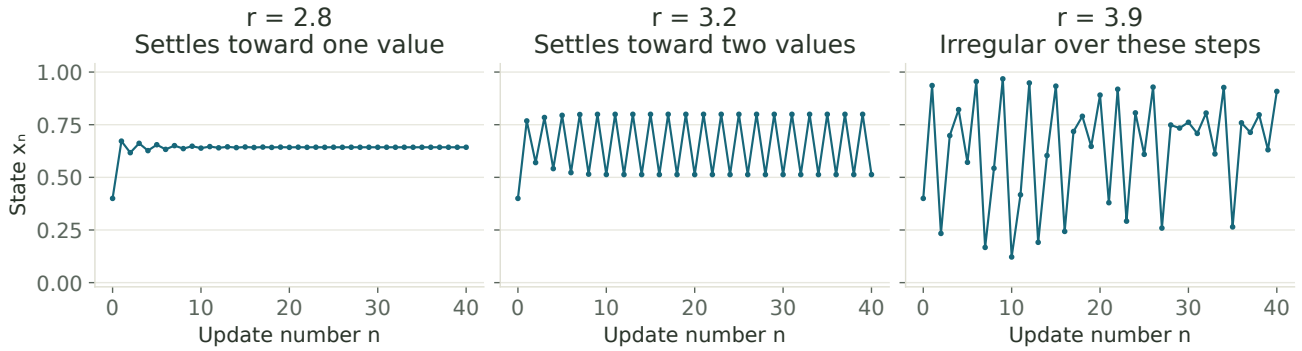
Three behaviors from the same formula

Now keep the initial state at 0.4 and change only r. Read each panel like the time graphs in Week 5: move left to right through the updates, and read the vertical value at each dot. Joining the dots helps the eye follow the sequence; the calculation itself happens at the numbered updates.

A *cycle* is a repeating sequence of states. A *two-cycle* alternates between two values, as a high-low-high-low pattern. The early values may still be settling toward that pattern. A nearly horizontal row of late dots indicates approach to a fixed point instead.

In Week 5, a feedback sequence could swing because of a strong correction or delayed information. This

rule uses the current population fraction directly at each season; it does not include a delayed sensor report. Here we investigate what happens as the response setting changes.



Each panel starts at $x_0 = 0.4$ and shows 40 updates. The parameter changes; the formula stays the same. A finite plot shows numerical behavior, not a proof about every future update.

Giving chaotic behavior a meaning

The third panel does not settle into the visible one-value or two-value pattern. Its states remain between 0 and 1 while moving irregularly. To investigate further, we will run the rule twice from almost identical starts.

Mathematicians use *chaos* for a kind of deterministic behavior in which very small initial differences can be amplified enough to limit detailed long-range forecasts. *Deterministic* means that an exactly specified input has one specified next result. In our chaotic logistic examples, later values also do not settle into a fixed point or a repeating cycle. This amplification of uncertainty about the start is called *sensitivity to initial conditions*. The comparison ahead shows what that means numerically.

Behavior	What to look for	Example
Fixed point	Later values approach one state.	r = 2.8 approaches $1 - 1/2.8 \approx 0.642857$.
Cycle	Later values approach a repeating pattern.	r = 3.2 approaches a two-cycle near 0.5130 and 0.7995.
Chaotic dynamics	Bounded irregular motion with sensitivity to nearby starts.	At r = 3.9, the nearby trajectories below separate substantially.

A larger r does not mean "more chaos" every time

For some ranges of r, repeating behavior returns. At $r = 3.83$, for example, ordinary nearby starts settle toward three values that repeat. Such a range is called a *periodic window*. The phrase simply means a range of parameter settings with repeating behavior.
There can also be special starting values. At $r = 4$, the exact start $x_0 = 0.5$ gives $0.5 \to 1 \to 0 \to 0$. These examples are why we study both the setting and the starting value, rather than labeling every run from the formula alone.

Pause and try 5

With $r = 2.8$ and $x_0 = 0.2$, calculate just the next state.

10.7 When do plausible forecasts become too different?

A conservation decision may tolerate small differences between population estimates but require a new survey when the predictions diverge too much. That acceptable difference is a question about the management goal, not a special number supplied by the formula.

Suppose a survey estimates 40,000 fish, but a count of 40,010 is also plausible. We have one lake and two possible descriptions of its initial state. Running the same model from both estimates lets us ask whether that small uncertainty matters later.

Using a reference population of 100,000 fish, those starts become $40{,}000/100{,}000 = 0.4000$ and $40{,}010/100{,}000 = 0.4001$. We call the two forecast sequences x_n and y_n, so that we can compare their values at matching update numbers.

A deterministic rule gives the same next state whenever its input is exactly the same. In practice, an observation usually identifies a range of possible starting states. We can follow two nearby starts to see whether their forecasts remain close.

Example · A difference of one ten-thousandth

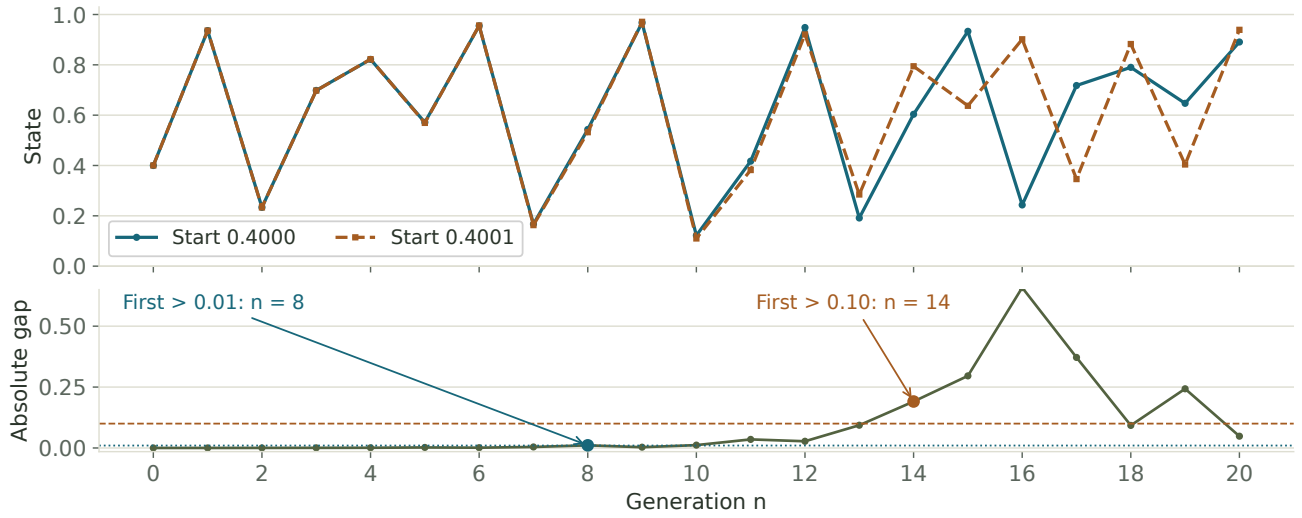
Use $r = 3.9$ with $x_0 = 0.4000$ and $y_0 = 0.4001$. The first updates are

$$x_1 = 3.9(0.4)(0.6) = 0.936, \qquad y_1 = 3.9(0.4001)(0.5999) = 0.936077961.$$

The initial gap is 0.0001; after the first update it is about 0.000078. The gap *shrinks initially*. Sensitivity does not require the gap to grow at every step.

Read the two panels as two different questions

In the upper panel, each line answers "what state does this starting estimate predict at generation n?" In the lower panel, each dot answers "how far apart are those two predictions at that same generation?"
The gap $|x_n - y_n|$ is found by subtracting the smaller state from the larger. It is zero when the forecasts agree. A gap of 0.01 means one hundredth of the reference population, or 1,000 fish in this setting. It does not mean 0.01 fish.



The upper plot shows the states; the lower plot shows their absolute gap. Both calculations use full stored precision. Displayed decimals are rounded only for reading.

Choose how much disagreement matters

A *threshold* is a chosen cutoff. Suppose the question is "when does the gap first become greater than 0.01?" Read down the gap column and stop at the first value above that cutoff:

Generation	Absolute gap	Is the gap greater than 0.01?
6	0.001174	No
7	0.004172	No
8	0.010896	Yes – this is the first crossing
9	0.003207	No – the forecasts have moved closer again

In this example, the earlier gaps at generations 0 through 5 are also below 0.01. "Greater than" means strictly above; a gap of exactly 0.01 would not cross this threshold. The comparison symbol in $|x_n - y_n| > 0.01$ is a compact way to ask the same question.

A *tolerance* is how much disagreement a decision can accept. For this pair, a tolerance of 0.01 is first exceeded at generation 8; a tolerance of 0.10 is first exceeded at generation 14. That first-crossing time gives a *forecast horizon for this pair of starts and this tolerance*: the first step when the predictions disagree more than the chosen limit. A useful tolerance should reflect the decision, not merely what is convenient to calculate. A more demanding tolerance can give a shorter useful horizon.

Two runs explore only two plausible starting estimates. An actual assessment would also need survey-error estimates, parameter uncertainty, and evidence that the model fits observed populations. An exact calculation from a poorly fitted rule is not necessarily a trustworthy forecast.

Round for display, not between updates

If 0.4001 is rounded to 0.400 before the experiment begins, the two starts become identical and their apparent difference disappears. During iteration, premature rounding changes the input to every

later calculation. Keep the calculator's stored value and round only the numbers you report.
For the first few updates, substitute the current value into the rule. For longer sequences, use the explorer's table. Keep full precision during the updates so that rounding does not change the modeled trajectory.

Deterministic does not mean easily forecast

The logistic rule contains no random-number generator. Its sensitivity concerns the consequences of imperfectly known initial conditions. Random sampling, incomplete models, and noisy measurements can also limit prediction, but they are distinct sources of uncertainty.

Pause and try 6

At generation 8, the gap is about 0.010896. At generation 9 it is about 0.003207. Has the first crossing of 0.01 moved to a later generation?

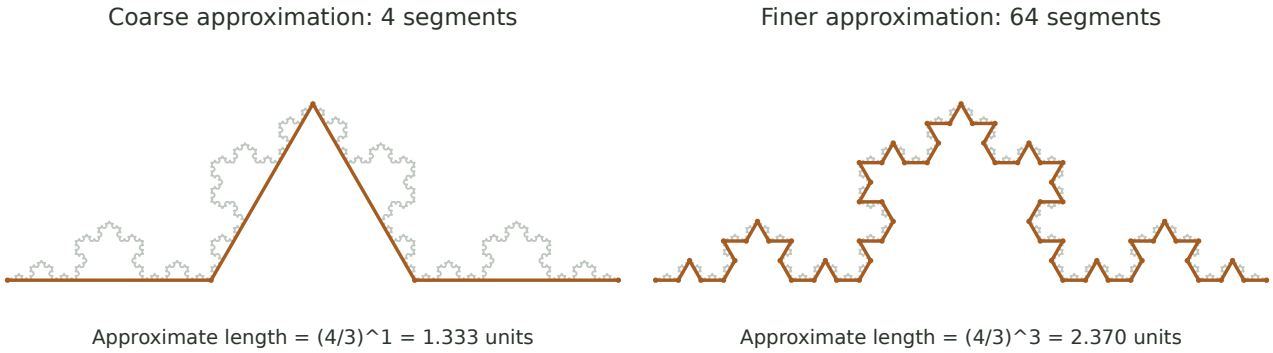
10.8 How long is the shoreline, at what scale?

Imagine planning a shoreline cleanup or habitat survey. A map adequate for choosing access points may leave out small bends that matter when field crews inspect the bank. The resolution of the observation is part of the measurement and should match the purpose.

Imagine laying a straight ruler along a winding shoreline on a map. Mark where one ruler length ends, move the ruler to begin there, and keep going. Every placement contributes one ruler length to the estimate. A long ruler can bridge across a small inlet; a shorter ruler can turn into it.

The ruler length sets the *measurement scale*. A finer scale follows smaller details. For these idealized tables, we treat every placement as a full ruler length, so multiplying the count by the ruler length gives the reported total.

A coarse map skips small bends in a boundary. A finer map follows more of them. The *coastline paradox* asks what happens to a reported length as the observation scale changes.



A schematic illustration, not field data. The gray boundary is the same in both panels. The colored measuring route changes with resolution.

Read the two pictures as two measurements of the same boundary. The gray detail stays in place. The colored straight pieces show the measuring routes: the finer route uses more, shorter pieces and follows more turns.

Example · State both the length and the ruler

In this invented survey, measured length equals ruler length times the number of placements.

Ruler length	Placements	Measured length	In kilometers
100 m	80	$100 \times 80 = 8{,}000$ m	8.00 km
50 m	200	$50 \times 200 = 10{,}000$ m	10.00 km
25 m	520	$25 \times 520 = 13{,}000$ m	13.00 km
12.5 m	1,340	$12.5 \times 1{,}340 = 16{,}750$ m	16.75 km

The shoreline has not grown because a smaller ruler was used. The reported length changes because the finer method follows detail the coarser method skips. A useful estimate for planning fieldwork includes the method, resolution, and a decision about which details matter.

In an ideal Koch curve, finite-stage lengths grow without bound. Natural coastlines are physical objects, with tides, changing shorelines, and limits to how a boundary is defined and observed. Approximate

fractal behavior across some scales does not establish literally infinite physical length.

Connection to optimization · Does a complicated boundary change the method?

In Week 8, straight-line constraints enclosed a polygon, and its corners gave a manageable list of candidate choices. A coastline is a useful reminder that the geometry we choose to represent a situation is part of the model. More detailed boundary data may require a different feasible-set description.

The corner-point method from Week 8 applies to a feasible polygon defined by linear inequalities. Replacing an edge with an irregular or fractal boundary changes the mathematical problem. A finite approximation may have many corners and may be nonconvex; the original list of vertices and linear constraints no longer describes it.

A bounded region can still have a finite, well-defined area even when its ideal boundary has infinite length. Infinite perimeter does not automatically imply undefined area. The chosen map resolution should match the decision we need to make.

Pause and try 7

A 20-meter ruler needs 600 placements. A 10-meter ruler needs 1,300 placements. Report both measured lengths.

10.9 Three limits of prediction, one habit of reasoning

The same mathematical habit has appeared in different settings. Week 2 asked what a distribution describes. Weeks 3 and 4 asked how data and uncertain measurements support a conclusion. Week 5 studied how the present affects the next update. Weeks 7 and 8 asked which choices a model permits and what "best" means. Week 9 made the connections themselves visible.

Three different limits have appeared. Uncertain starting measurements and incomplete models constrain useful forecasts. A finite computation does not prove a claim about every possible case. And a measurement can change when its scale changes. The table below separates these conclusions.

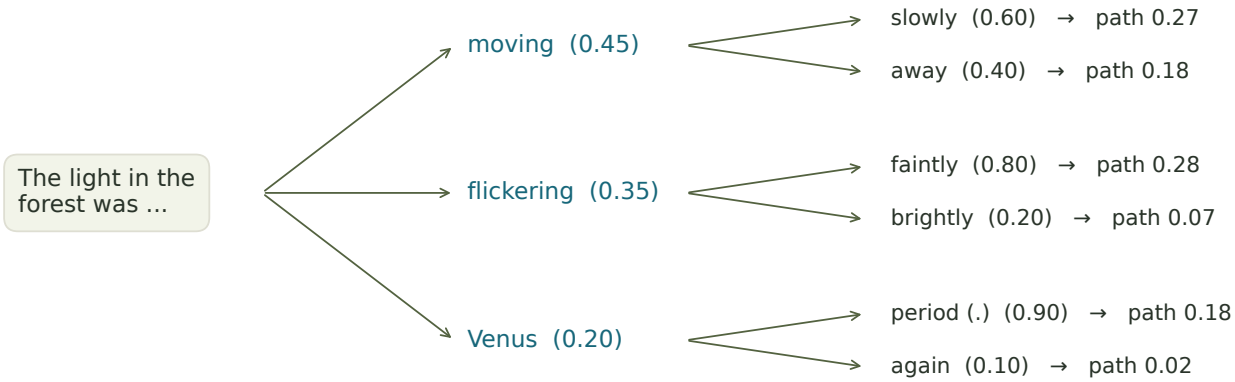
Example	What we can calculate	What still needs care
Collatz	The route, update count, and peak for a chosen start.	A finite collection of successes does not prove a universal claim.
Koch and Sierpinski	Exact finite-stage counts, lengths, and area fractions.	The drawn stage is finite; the ideal fractal is a limit.
Logistic map	The next state from an exact input; comparisons across starts.	Measurement uncertainty and sensitivity limit useful horizons.
Coastline	Length under a specified scale and method.	Different resolutions answer different measurement questions.
AI text generation	Probabilities in a model and the consequences of a chosen token.	A probable continuation is not evidence that its claim is true.

Across this course, we have measured, modeled, calculated, compared, and chosen. The habit to carry forward is to ask: *What do I know, how did I obtain it, and what decision does it support?* Mathematics makes conclusions precise; good reasoning also makes their limits precise.

The AI connection: the output becomes the next input

A language model uses its current text context to predict a next token, appends the chosen token, and then repeats with the updated context. A token is a piece of text; a probability distribution gives the weights of possible next pieces. This repeated process is called autoregressive generation.

The invented tree below treats each listed word or punctuation mark as a toy token. Begin at the prompt and follow one arrow at a time. Each second set of probabilities belongs to the first token already chosen.



For a relative-frequency picture, imagine 100 hypothetical runs divided in the stated proportions. Of the 45 that choose "moving," 60% then choose "slowly": $45(0.60) = 27$ out of the original 100, giving a path probability of 0.27. The path "flickering then faintly" has probability $0.35(0.80) = 0.28$.

This is why we multiply along a path. The notation $P(\text{slowly} \mid \text{moving})$ means the probability of "slowly" *given* that "moving" was chosen. A likely continuation still needs observations or sources to establish whether a claim about an actual light is true. The online AI application develops this example with a selectable tree.

Chapter Review

1. Follow Collatz from 11. Record its update count and peak. If another trajectory later reaches 11, what happens to its remaining route?
2. At Koch stage 4, with an original segment of length 1, find the segment count, length per segment, and total length. How does the total change if the original length is 9 meters?
3. At Sierpinski stage 4, find the filled-triangle count and remaining area fraction. How can these move in different directions?
4. For $r = 2.8$, start at 0.2. Calculate the first update and the nonzero fixed point. Why are they different quantities?
5. For $r = 3.8$ with starts 0.40 and 0.41, calculate the first update at which the absolute gap exceeds 0.05. Retain full stored precision between updates.
6. In the $r = 3.9$ example, let a model population be $N_n = 100{,}000x_n$ fish. How many fish do the starting states 0.4000 and 0.4001 represent? At which generations does their gap first exceed 1,000 fish and 10,000 fish?
7. A 50-meter ruler requires 260 placements and a 25-meter ruler requires 580. Report both measured coastline lengths with their scales.
8. A computer verifies Collatz for every integer in a finite interval. What conclusion is justified? Give a family of infinitely many starting integers whose convergence can be proved directly.
9. In a toy AI model, $P(\text{moving}) = 0.45$ and $P(\text{slowly} \mid \text{moving}) = 0.60$, while $P(\text{flickering}) = 0.35$ and $P(\text{faintly} \mid \text{flickering}) = 0.80$. Compare the two path probabilities. Does either probability establish what an actual light in a forest was doing?

Optional connection / A boundary can grow while area stays finite

The following area calculation is optional. For an equilateral triangle with side length s, you may use $A = (\sqrt{3}/4)s^2$ without deriving it. The main point is to compare what happens to length with what happens to area.

Begin with a rectangle 10 units wide and 5 units high, of area 50. Replace its top edge by outward-pointing Koch bumps. At stage 1, the added triangle has side $10/3$ and area $25\sqrt{3}/9 \approx 4.8113$. At stage 2, four triangles of side $10/9$ add $100\sqrt{3}/81 \approx 2.1383$. Thus the area is approximately 56.9496, while the perimeter is $20 + 10(4/3)^2 \approx 37.7778$.

At later stages, each added area is $4/9$ of the preceding addition, so their sum converges while the top-edge length diverges. Infinite ideal boundary length does not imply undefined area. Such a boundary also changes the feasible-set model; the original linear inequalities and corner list no longer describe it.

Solutions and Further Reading

Check 1. The values become 32 and 32.8. The absolute gap is 0.8. Relative to 32, the gap is $0.8/32 = 0.025 = 2.5\%$, exactly the initial relative gap.

Check 2. $5 \to 16 \to 8 \to 4 \to 2 \to 1$: 5 updates, peak 16. Count arrows, not recorded numbers.

Check 3. The conjecture holds for those 10,000 starting integers. The experiment provides evidence for the universal conjecture; it does not prove it for all positive integers.

Check 4. Each segment has length $(1/3)^3 = 1/27$. There are 64 segments, so $L_3 = 64/27$. The one-third factor is applied three times.

Check 5. $x_1 = 2.8(0.2)(0.8) = 0.448$. It is not the fixed point; it is the first update from the chosen start.

Check 6. No. The first crossing is still generation 8. The trajectories can later move closer again. "First exceeds" does not mean "stays above forever."

Check 7. The coarse measurement is $20 \cdot 600 = 12{,}000$ meters (12 km). The finer measurement is $10 \cdot 1{,}300 = 13{,}000$ meters (13 km). Report the ruler sizes along with the lengths.

Review answers

1. From 11: 14 updates, peak 52. Routes that reach 11 follow its future route.
2. 256 segments, each of length 1/81, total $256/81 \approx 3.1605$. From a 9-meter original, total length is $9(256/81) = 256/9 \approx 28.4444$ meters.
3. 81 triangles; area $81/256 \approx 0.3164$. Each new triangle has one-quarter its parent's area; the three remaining pieces together retain only three quarters.
4. First update 0.448; nonzero fixed point $1 - 1/2.8 \approx 0.642857$. The first depends on the chosen start; the second is unchanged by the update.
5. Update 4, with gap approximately 0.078197. The preceding three gaps are about 0.007220, 0.022805, and 0.035779.
6. 40,000 and 40,010 fish; first gaps above 1,000 and 10,000 fish occur at generations 8 and 14. At generation 14, the gap is approximately 19,074 fish. The reference population is a scale chosen for this toy model, rather than a fitted carrying capacity for a real lake.
7. 13 km using the 50-meter ruler; 14.5 km using the 25-meter ruler.
8. The tested interval only. Every power of 2 reaches 1 by repeated halving.
9. $0.45(0.60) = 0.27 < 0.35(0.80) = 0.28$. Model continuation probabilities do not verify claims about the world.

Further reading

These sources provide background, not additional required assignments.

- R. M. May (1976), *Simple mathematical models with very complicated dynamics*. Nature 261, 459–467.
- R. De Leo and J. A. Yorke (2021), *The graph of the logistic map is a tower*. DCDS 41, 5243–5269.
- A.-S. Elsenhans (2026), *Numerical verification of the Collatz conjecture for billion digit random numbers*.
- B. Mandelbrot (1967), *How long is the coast of Britain?* Science 156, 636–638.
- A. Vaswani et al. (2017), *Attention Is All You Need*.